

Open NeuroHealth Framework (ONF)

A Global Open Framework for Neuroscience, Digital Brain Health, and Research Collaboration

Executive Summary

The **Open NeuroHealth Framework (ONF)** is a unified, ethical, and open platform connecting early neurological diagnosis, patient care, neuroscience research, and education. It aims to make brain health accessible, measurable, and improvable for every human through open digital neuroscience.

Core Outcomes: - Global early-stroke detection via smartphones. - Integration of clinical and research data for neurovascular and blood-brain barrier (BBB) science. - A collaborative ecosystem for AI, education, and neuro-innovation.

Problem Statement

Global Burden of Neurological Disease

Stroke and neurodegenerative disorders remain the leading causes of disability worldwide. Over 70% of the population lacks access to early diagnostics.

Gaps in Current Systems

Gap	Consequence
Fragmented diagnostics	Missed early signs, delayed treatment
Isolated research silos	Repetition, slow innovation
Lack of data diversity	Biased AI models
Weak neuroscience education	Low public awareness
Closed technology	Limits collaboration

Why ONF is Needed

Neuroscience lacks a standardized, interoperable, and ethical digital layer comparable to DICOM for imaging or HL7 for health data. ONF fills this gap and provides a framework for scalable, transparent, and equitable neuro-health innovation.

Proposed Solution – The ONF Architecture

Four Functional Pillars:

Pillar	Description	Example
NeuroHealth	Early detection and continuous monitoring	AI-based stroke module
NeuroCare	Clinical dashboards and treatment tracking	Tele-neuro integration
NeuroResearch	Open, de-identified datasets and ML sandboxes	BBB dataset for labs
NeuroEducation	Digital learning and citizen-science platform	MOOCs and simulation tools

Interoperability Layer

- REST APIs + JSON schema for standardized **Neuro Units**.
- Plug-in SDKs for Python, R, and MATLAB.

Data Governance

- Local data ownership with globally aggregated, anonymized insights.
- Automated consent tracking, anonymization, and bias audits.

Neuro Unit Specification

Definition: The Neuro Unit is the basic data object of the ONF, representing a clinical, research, educational, or therapeutic contribution.

Type	Example Data	Exchange Method
Clinical Unit	MRI scan + symptoms	Hospital → ONF Cloud
Research Unit	EEG dataset + code	Lab → Repository
Education Unit	Simulation / lecture	Professor → ONF Portal
Treatment Unit	Rehab protocol	Clinician → Care Library

Each unit contains metadata, DOI, ethics code, and a reusability certificate for open-science traceability.

Implementation Roadmap

Phase	Years	Key Deliverables
I. Prototype Era	0–3	AI Stroke App, first API draft, pilot hospitals

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II. Platform Era	3–6	ONF OS, SDK release, developer portal
III. Ecosystem Era	6–10	Research Cloud, Education Portal, standard v2.0
IV. Infrastructure Era	10–20	WHO/UNESCO adoption, Global Consortium

Governance and Licensing Model

Structure

- **ONF Foundation (non-profit)** – owns open standards and ethics.
- **NeuroHealth Technologies (for-profit)** – runs operations and licensing.

Licensing

Use Case	License Type
Academic	Free, CC-BY-NC-SA
Commercial	Revenue-share or partnership
Government / WHO	Free for humanitarian use

Open Science Charter

- Transparency
- Reproducibility
- Privacy
- Non-discrimination
- Global accessibility

Partnerships

- **Universities:** IISc, NBRC, Amrita, NUS
 - **Hospitals:** AIIMS, NIMHANS, Stanford Health
 - **Industry:** Google Health, Philips, Siemens
 - **NGOs:** WHO, UNESCO, Gates Foundation
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Potential Impact

Domain	10-Year Impact	20-Year Vision
Clinical	25% drop in stroke-related disability	Global neuro-prevention system
Research	First open BBB dataset	Predictive neuroscience revolution
Education	1M trained students	New neuro-engineering generation
Society	Improved neuro-literacy	Brain health = public health
Economy	\$50B+ in global savings	Sustainable neuro-economy

Call to Action

"We invite researchers, clinicians, educators, and innovators to join the **Open NeuroHealth Framework** — a collective mission to make brain health measurable, accessible, and improvable for every human being."

Contact: [To be added]

GitHub: [To be added]

Website: [To be added]
